

evaluateCD.py contain useful codes for KNN and analogy tasks evaluation
simlex-999 golden standard templateLLY.py is a sample code for simlex-999 golden standard evaluation

Project 1:

Train and evaluate the Continuous Bag of Words (CBOW) model.

I. Prerequisite: Installing Anaconda, Pytorch, and nltk library.

This part is an instruction for installing the prerequisites for running the program. It is for your own reference and will not be covered in the lecture.

Please note that if you have already installed and used Pytorch successfully on your computer, just skip part 1 and 2.

1. Anaconda

Download the Installer from the official website download link:

<https://www.anaconda.com/>

If cannot open it, turn to the open source software mirror station of Tsinghua University:

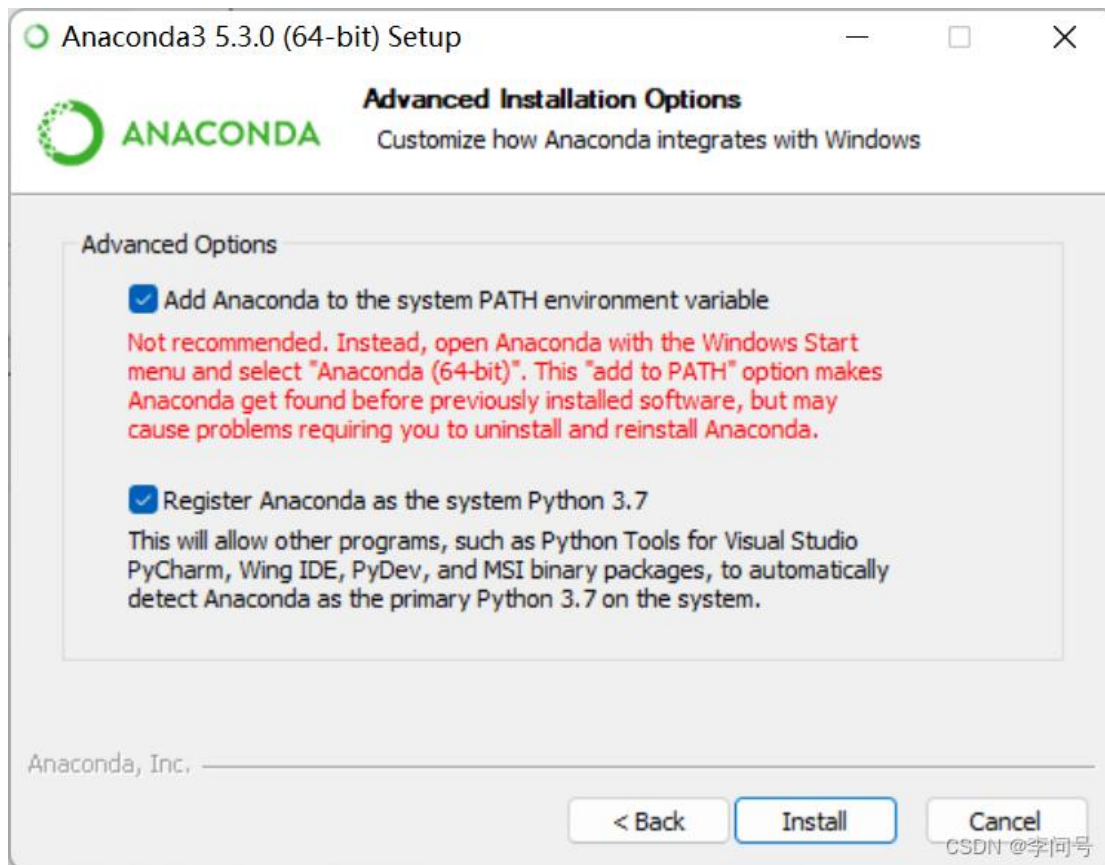
<https://mirrors.tuna.tsinghua.edu.cn/anaconda/archive/>

Then choose among the newest versions according to the OS and CPU type of your computer.

Anaconda3-2023.09-0-Linux-aarch64.sh	838.8 MiB	2023-09-29 23:47
Anaconda3-2023.09-0-Linux-ppc64le.sh	525.2 MiB	2023-09-29 23:47
Anaconda3-2023.09-0-Linux-s390x.sh	366.2 MiB	2023-09-29 23:47
Anaconda3-2023.09-0-Linux-x86_64.sh	1.1 GiB	2023-09-29 23:47
Anaconda3-2023.09-0-MacOSX-arm64.pkg	741.8 MiB	2023-09-29 23:47
Anaconda3-2023.09-0-MacOSX-arm64.sh	744.0 MiB	2023-09-29 23:47
Anaconda3-2023.09-0-MacOSX-x86_64.pkg	772.0 MiB	2023-09-29 23:48
Anaconda3-2023.09-0-MacOSX-x86_64.sh	774.1 MiB	2023-09-29 23:48
Anaconda3-2023.09-0-Windows-x86_64.exe	1.0 GiB	2023-09-29 23:48

For example, if your computer have windows system and Intel x86 CPU, then choose the last one shown in the picture above.

After downloading, open the installer and follow its instructions to install. You can always click “next” for each step, except that you should check both boxes in this step so that Anaconda will be automatically added to the system path, or you will have to add it by yourself.



Finally, you can check whether the installation was successful by typing **conda --version** on the command line.

```
C:\Windows\system32\cmd.exe
Microsoft Windows [版本 10.0.22000.978]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\peace>conda --version
conda 4.12.0

C:\Users\peace>
```

2. Pytorch

First, go to the official website: <https://pytorch.org/>

Find the installing options in the website, Adjust the options according to your needs, and then copy the installation command generated at the bottom into the command line and run it. Remember to type 'y' when the command line asks for your permission.

PyTorch Build	Stable (2.1.0)		Preview (Nightly)	
Your OS	Linux	Mac	Windows	
Package	Conda	Pip	LibTorch	Source
Language	Python		C++ / Java	
Compute Platform	CUDA 11.8	CUDA 12.1	ROCm 5.6	CPU
Run this Command:	<pre>pip3 install torch torchvision torchaudio</pre>			

Wait until the installation completes, then you can use Pytorch to run program for this task.

3. nltk

In this task, we need to use the reuters dataset in the nltk library to train CBOW word vectors.

First, type **pip install nltk** in the command line to download the library.

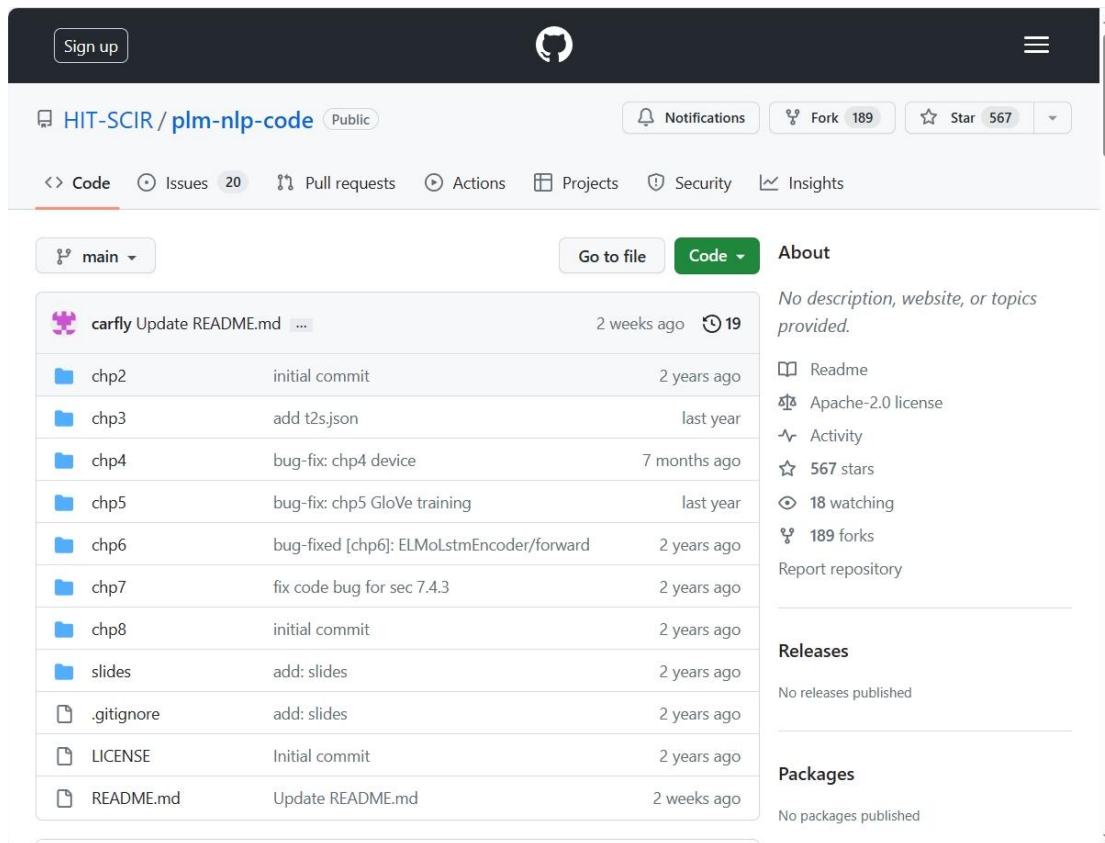
Then, run the following code in Python

```
import nltk
nltk.download("reuters")
```

to download the reuters dataset.

II. Download and run the code for CBOW

First, open this website: <https://github.com/HIT-SCIR/plm-nlp-code>



Click the green code button in the upper right corner, then click "Download ZIP" to download the zip file.

After the downloading is complete, unzip the compressed file and you will see the following files. Only the files in the chp5 folder will be used in this task, so you can drop other files or save them for later use.

名称	修改日期	类型	大小
chp2	2023/10/28 9:17	文件夹	
chp3	2023/10/28 9:17	文件夹	
chp4	2023/10/28 9:17	文件夹	
chp5	2023/11/9 15:20	文件夹	
chp6	2023/10/28 9:17	文件夹	
chp7	2023/10/28 9:17	文件夹	
chp8	2023/10/28 9:17	文件夹	
slides	2023/10/28 9:17	文件夹	
.gitignore	2023/10/28 9:17	Git Ignore 源文件	1 k
LICENSE	2023/10/28 9:17	文件	12 k
README.md	2023/10/28 9:17	Markdown 源文件	6 k

Open cbow.py file in chp5 folder and run the code in it. This program will run 10 epochs to learn CBOW word vectors. Each epoch may take about an hour to run on the CPU. If you want to quickly test the function of the program, you can set the global variable num_epoch to a smaller value.

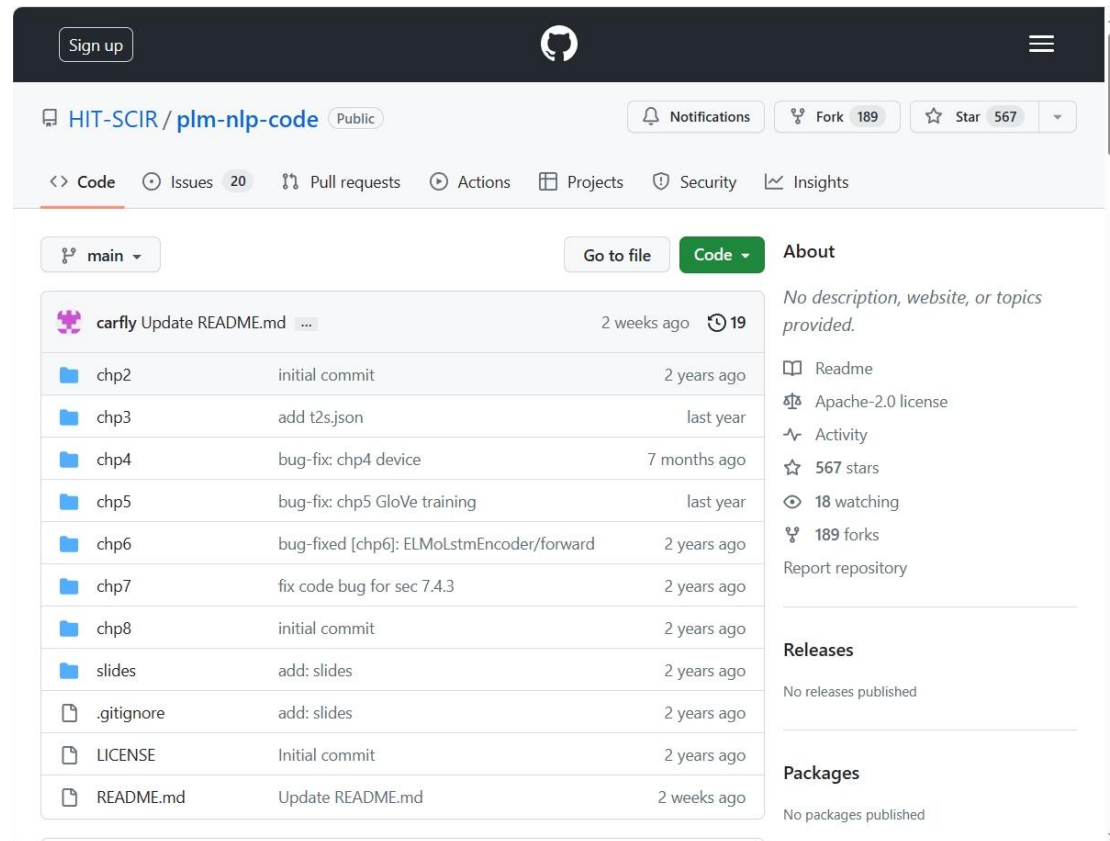
After the program completes its running, the word list used by CBOW and the generated word vectors will be saved in text form to the cbow.vec file in the same folder in which the first line specifies the vocabulary size and embedding dimension followed by all token and their vectors, one token per line (in NLP, a more generalized name for a word is "a token"), and at the top of them are some special added tokens and there is no need to be surprised by them.

Project 2

Train the Skip-gram Model

I. Source Code Download (The same as CBOW; skip if you have already done so)

Source code link: <https://github.com/HIT-SCIR/plm-nlp-code>



After downloading and unzipping:

名称	修改日期	类型	大小
chp2	2023/10/28 9:17	文件夹	
chp3	2023/10/28 9:17	文件夹	
chp4	2023/10/28 9:17	文件夹	
chp5	2023/11/9 15:20	文件夹	
chp6	2023/10/28 9:17	文件夹	
chp7	2023/10/28 9:17	文件夹	
chp8	2023/10/28 9:17	文件夹	
slides	2023/10/28 9:17	文件夹	
.gitignore	2023/10/28 9:17	Git Ignore 源文件	1 k
LICENSE	2023/10/28 9:17	文件	12 k
README.md	2023/10/28 9:17	Markdown 源文件	6 k

II. Training the Models

1. Skip-gram

(a)

The source file for ordinary Skip-gram is `skipgram.py` in the `chp5` folder. The trained word embedding from this file will be saved to a `.vec` file with the same name in text form; the same applies to the following.

(b)

Negative sampling converts the multi-classification task of predicting the correct word into the binary classification task of judging whether a word appears in the context, thereby greatly reducing the computational load.

The source file for Skip-gram with negative sampling is `sgns.py` in the `chp5` folder.

III. Evaluating the Model

The evaluation methods are same as for CBOW model.